CS540 – Principles of Artificial Intelligence

Course Details

- Begin and end: May 20 to August 26, 2026
- Instructor: Marcus Birkenkrahe, PhD
- Date and time: Wednesday, 6-7:30 pm PDT (class meeting)
- Asynchronous: 90 min (DataCamp platform)

Course Description

Principles of Artificial Intelligence provides a practical and conceptual grounding in modern machine learning and AI systems. The course emphasizes modeling, implementation, and evaluation of algorithms that operate on real-world data and produce actionable outputs. Students will engage with supervised, unsupervised, and deep learning methods, culminating in an introduction to reinforcement learning.

The course integrates:

- Conceptual understanding (from Géron)
- Hands-on implementation (DataCamp + coding)
- Critical evaluation of models and assumptions

Learning Objectives

By the end of the course, students will be able to:

1. Explain core machine learning paradigms (supervised, unsupervised, reinforcement learning)
2. Implement regression and classification models
3. Evaluate model performance using appropriate metrics
4. Apply tree-based and ensemble methods
5. Perform clustering and dimensionality reduction
6. Build and train neural networks and CNNs
7. Describe and implement basic reinforcement learning methods
8. Analyze limitations, assumptions, and failure modes of AI systems

Prerequisites

Students are expected to have:

- Basic programming skills (functions, loops, data structures)
- Introductory algebra
- Familiarity with arrays

If these are missing, students must complete preparatory material before Week 1.

Course Materials

- Aurélien Géron, *Hands-On Machine Learning* with scikit-learn and PyTorch (2026) - see GitHub.
- DataCamp (assigned lessons weekly)

- Instructor-provided notes and exercises

Course Structure

Each week consists of:

- Conceptual lecture (theory, assumptions, limitations)
- Guided coding (in-class or demonstration)
- DataCamp lesson (hands-on reinforcement)
- Practice problem or mini-task

Grading

- Weekly exercises / participation: 25%
- Midterm project (model + analysis): 25%
- Final project (end-to-end ML workflow): 25%
- Quizzes / checks: 25%

DataCamp / Textbook Alignment

Week	Géron Chapter	Topic	DataCamp Course	Chapter	Prereq
1	1: The ML Landscape	What is Machine Learning	Colab with Python assignment (not DataCamp)	1	—
2	2: End-to-End ML Project	ML Workflow	Supervised Learning with scikit-learn	1	Week 1
3	3: Classification	Classification (kNN)	Supervised Learning with scikit-learn	2	Week 2
4	4: Training Models	Linear + Logistic Regression	Supervised Learning with scikit-learn	3	Week 3
5	4 cont.	Model Evaluation / CV	Supervised Learning with scikit-learn	4	Week 4
6	5: Decision Trees	Decision Trees	Machine Learning with Tree-Based Models in Python	1	Week 5
7	6: Ensemble Learning	Bagging / Random Forests	Machine Learning with Tree-Based Models in Python	2–3	Week 6
8	6 cont.	Boosting / Model Tuning	Machine Learning with Tree-Based Models in Python	4–5	Week 7
9	7: Dimensionality Reduction	PCA	Unsupervised Learning in Python	3	Week 5
10	8: Unsupervised Learning	Clustering (k-means)	Unsupervised Learning in Python		
11	9: Intro to ANNs	Neural Networks (basics)	Introduction to Deep Learning with PyTorch	1	Week 5

Week	Géron Chapter	Topic	DataCamp Course	Chapter	Prereq
12	10: Building NNs with PyTorch	PyTorch Fundamentals	Introduction to Deep Learning with PyTorch	2–3	Week 11
13	11: Training Deep NNs	Training Neural Networks	Intermediate Deep Learning with PyTorch	1–2	Week 12
14	12: Deep Computer Vision	Convolutional Neural Networks	Deep Learning for Images with PyTorch	1–2	Week 13
15	18: Reinforcement Learning	RL / Q-learning	Reinforcement Learning with Gymnasium in Python	1–2	Week 5

Verification and Correctness Checks

Students will be expected to:

- Reproduce model results on small datasets
- Compare models using evaluation metrics
- Identify overfitting/underfitting
- Explain model behavior qualitatively

AI Usage Guidelines

The use of AI tools (including ChatGPT, Copilot, DataCamp assistants, and similar systems) is permitted but regulated. The goal is to use AI as a **tool for learning and productivity**, not as a substitute for understanding.

Permitted Uses

Students may use AI tools to:

- Clarify concepts encountered in lectures or readings
- Debug code after attempting a solution independently
- Generate alternative explanations or examples
- Explore extensions or variations of assigned problems

Restricted Uses

Students may **not**:

- Submit AI-generated code or text as their own without modification and understanding
- Use AI to bypass problem-solving (e.g., directly generating full solutions)
- Rely on AI outputs without verification

Required Practices

When using AI tools, students must:

1. Be able to explain any code or solution they submit
2. Verify outputs independently (e.g., by testing or reasoning)
3. Treat AI outputs as **hypotheses**, not authoritative answers

Transparency Requirement

For major assignments and projects:

- Students must include a brief note describing any AI assistance used
- Example:

  > Used ChatGPT to clarify how cross-validation works and to debug a shape mismatch error.
  > All code was written and verified independently.

Evaluation Policy

- Understanding takes precedence over correctness
- Students may be asked to explain or reproduce their work
- Inability to explain submitted work may result in reduced credit

Rationale

AI tools are powerful but imperfect:

- They may produce incorrect or misleading outputs
- They do not replace conceptual understanding
- Effective use requires critical evaluation

The course treats AI as:

- A **tool** for exploration and productivity
- A **subject** of critical analysis

CPU policies

For policy matters and procedures, especially concerning attendance, grading tables, withdrawals etc. please consult the Catholic Polytechnic University's [CATALOG OF COURSES (2025-2026)](#).

Date: Summer Term - Catholic Polytechnic University (Graduate Program)
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